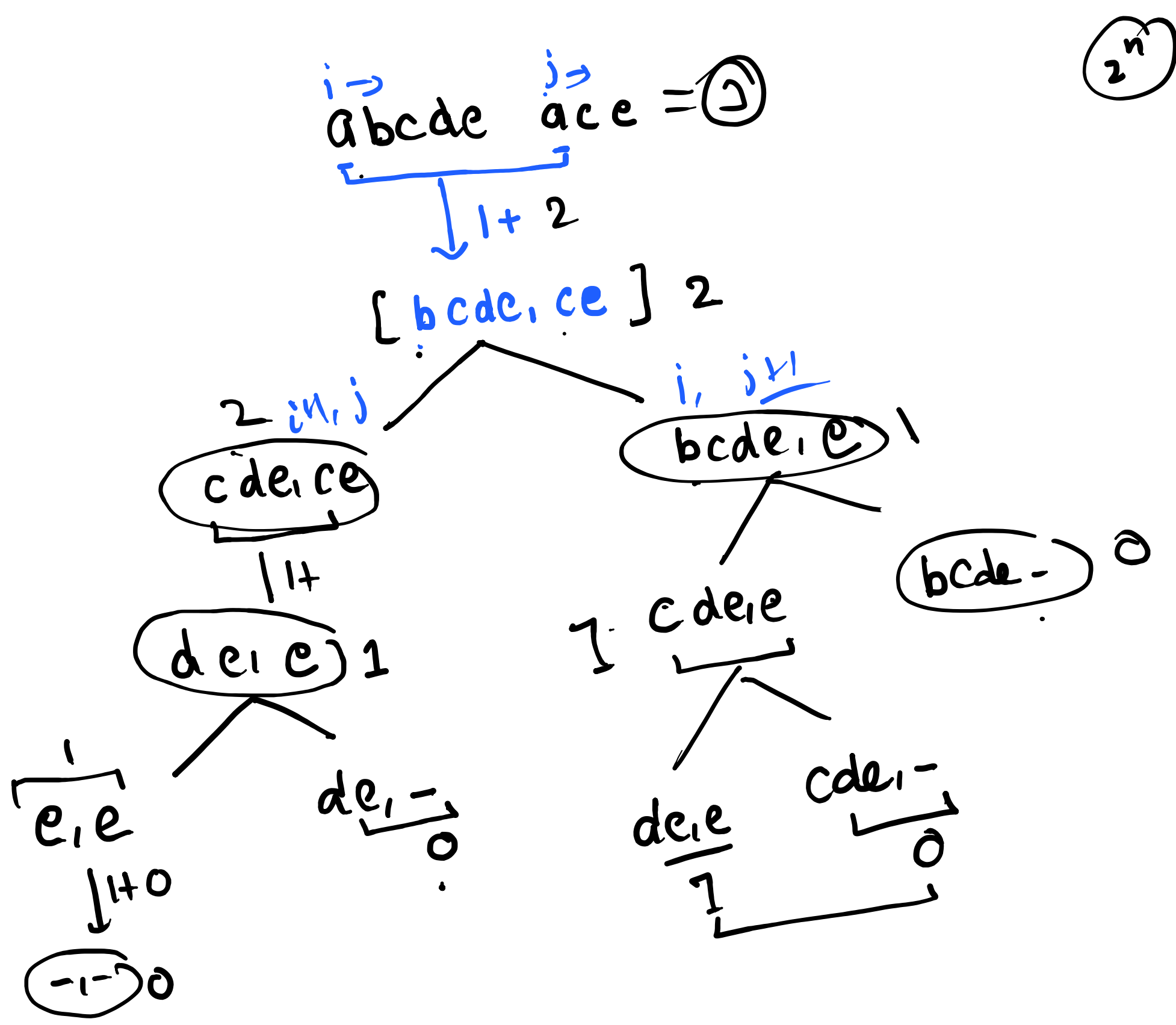
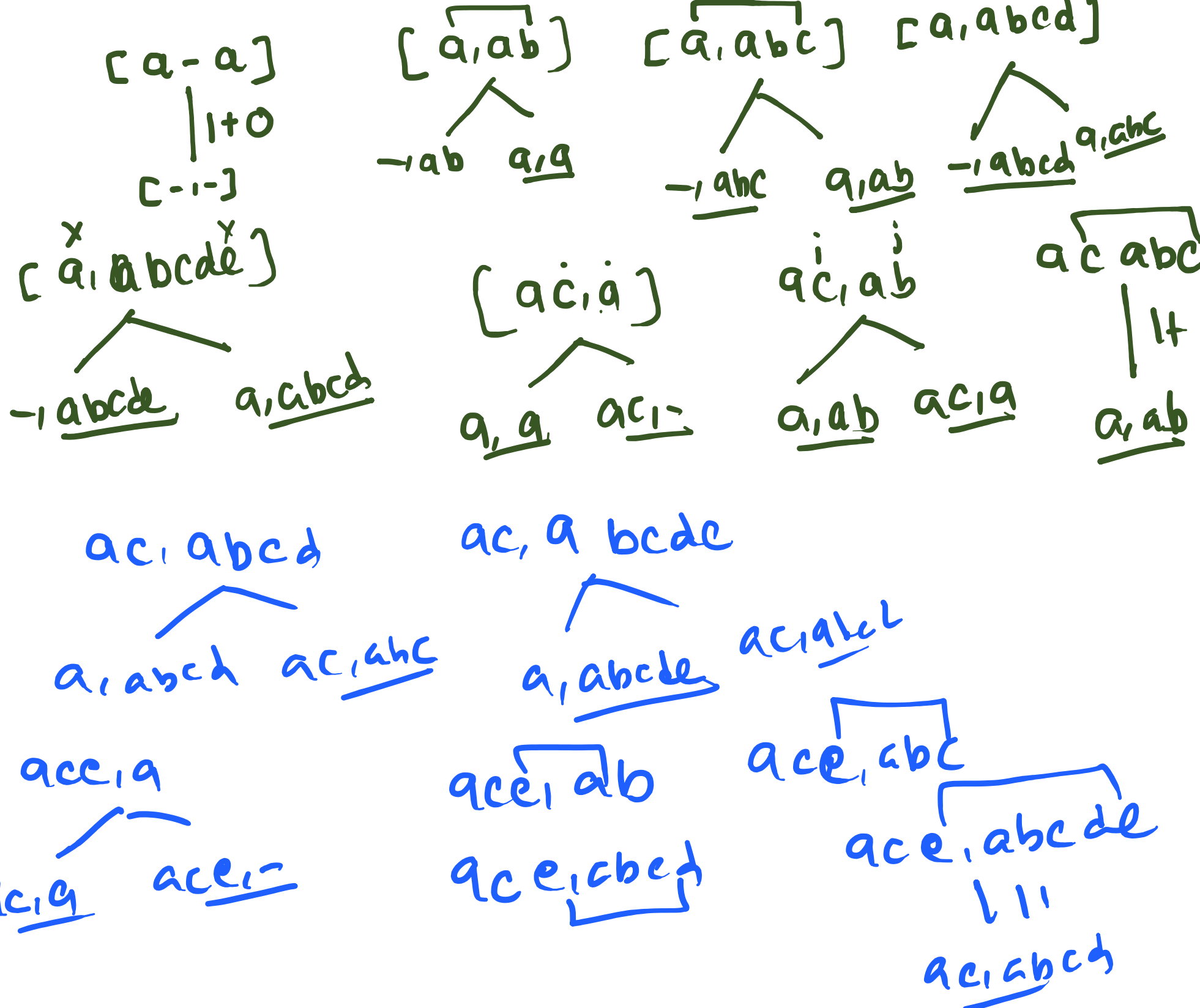




	-	a	b	c	d	e
-	-	-1a	-1ab	-1abc	-1abd	-1abce
a	a-	0	1	1	1	1
c	ac-	0	0	1	1	1
e	ace-	0	0	0	0	1

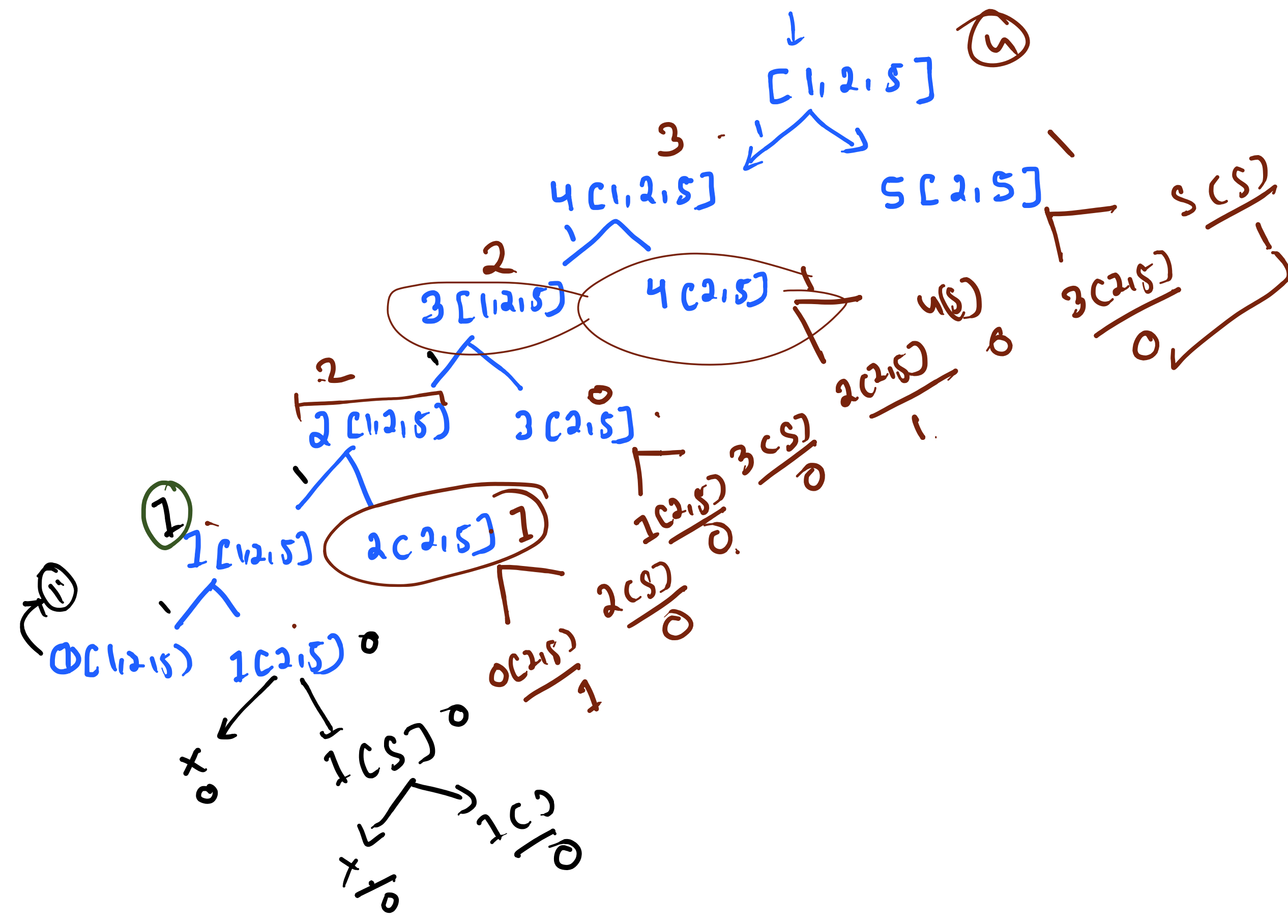
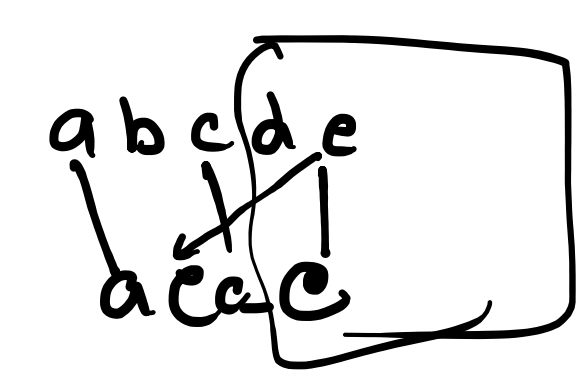
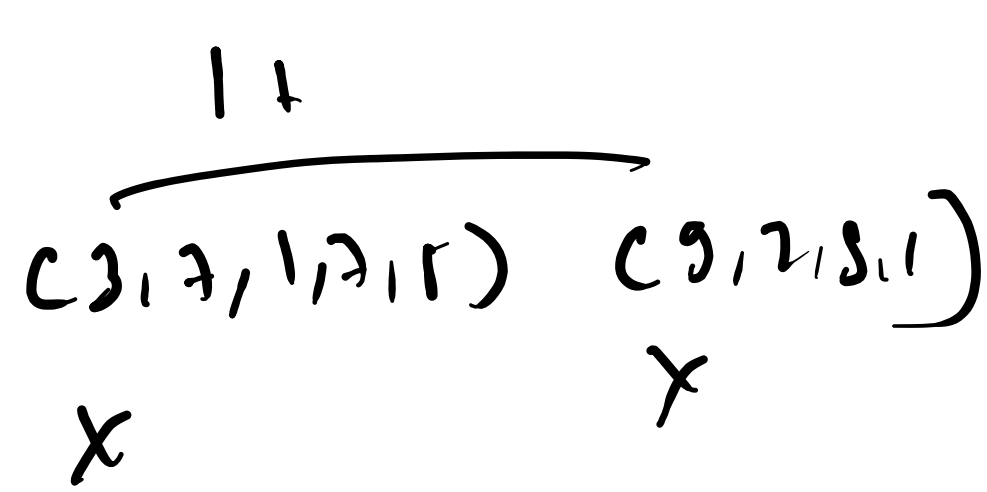
	-	a	b	c	d	e
-	0	0	0	0	0	0
a	0	1	1	1	1	1
c	0	1	1	2	2	2
e	0	1	1	2	2	3

Handwritten formulas for calculating LCS length using dynamic programming:

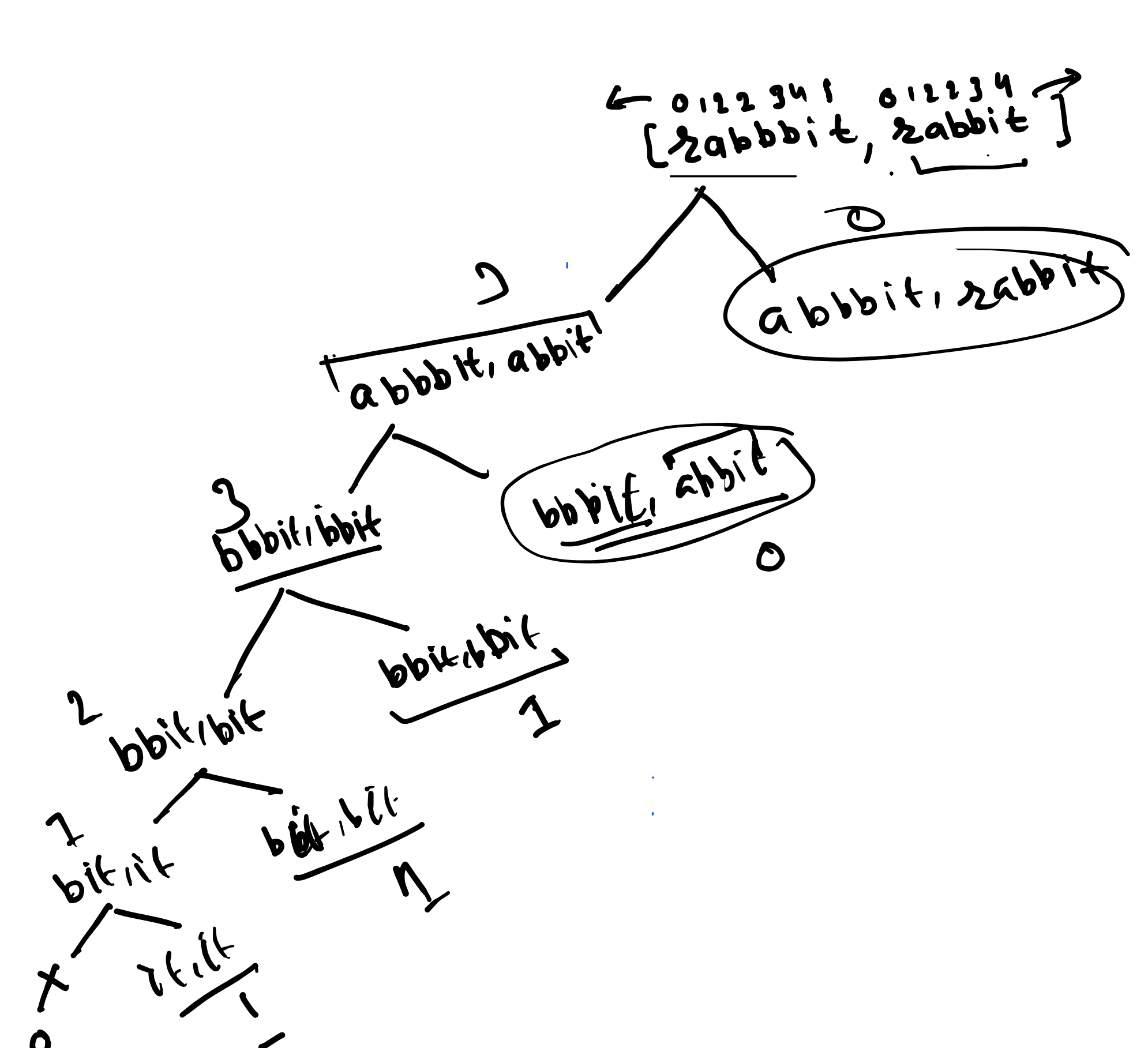
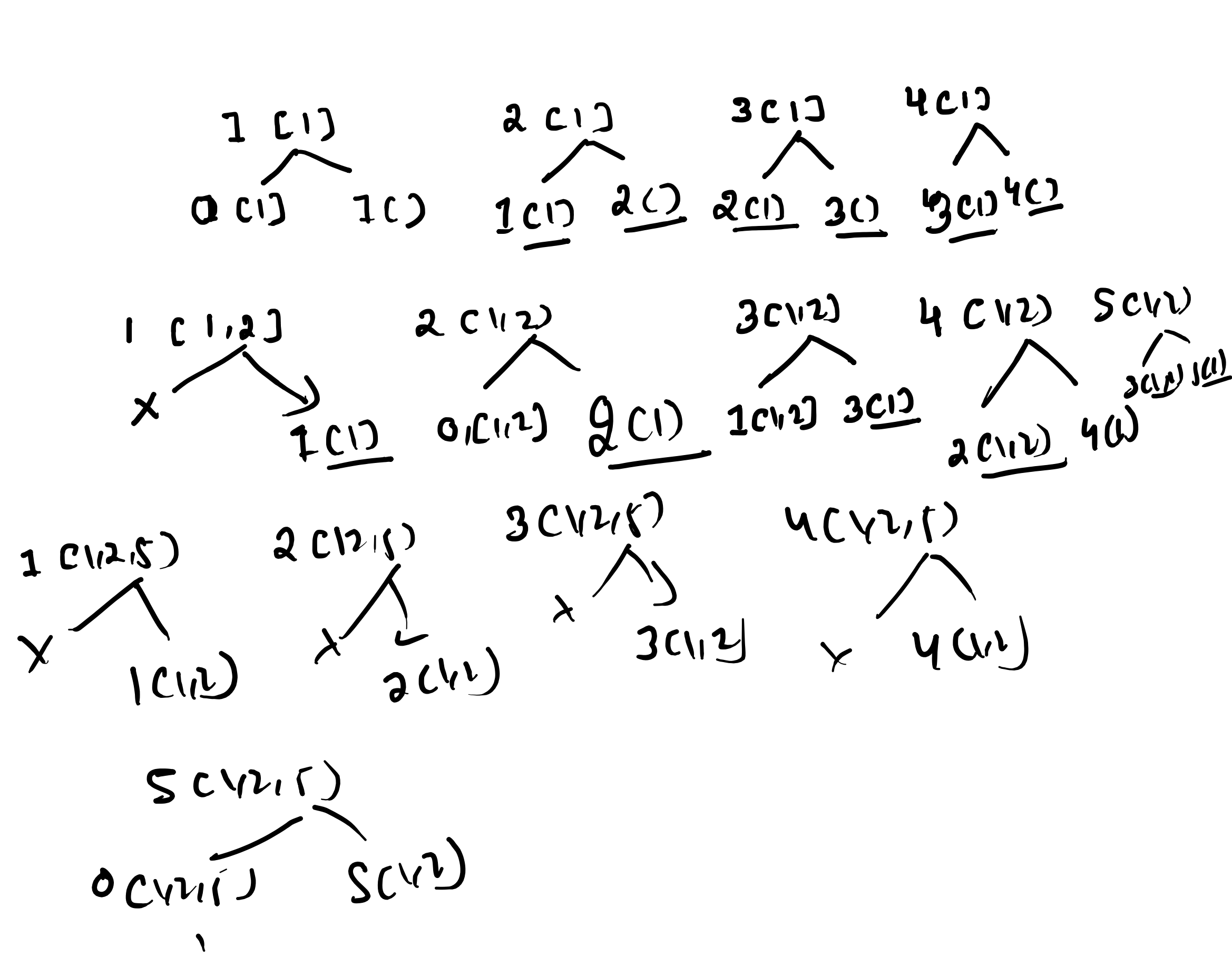
$$dp[i][j] = \begin{cases} dp[i-1][j-1] + 1 & \text{if } s[i] == t[j] \\ \max(dp[i-1][j], dp[i][j-1]) & \text{otherwise} \end{cases}$$


nums1 = [1,3,7,1,7,5], nums2 = [1,9,2,5,1]

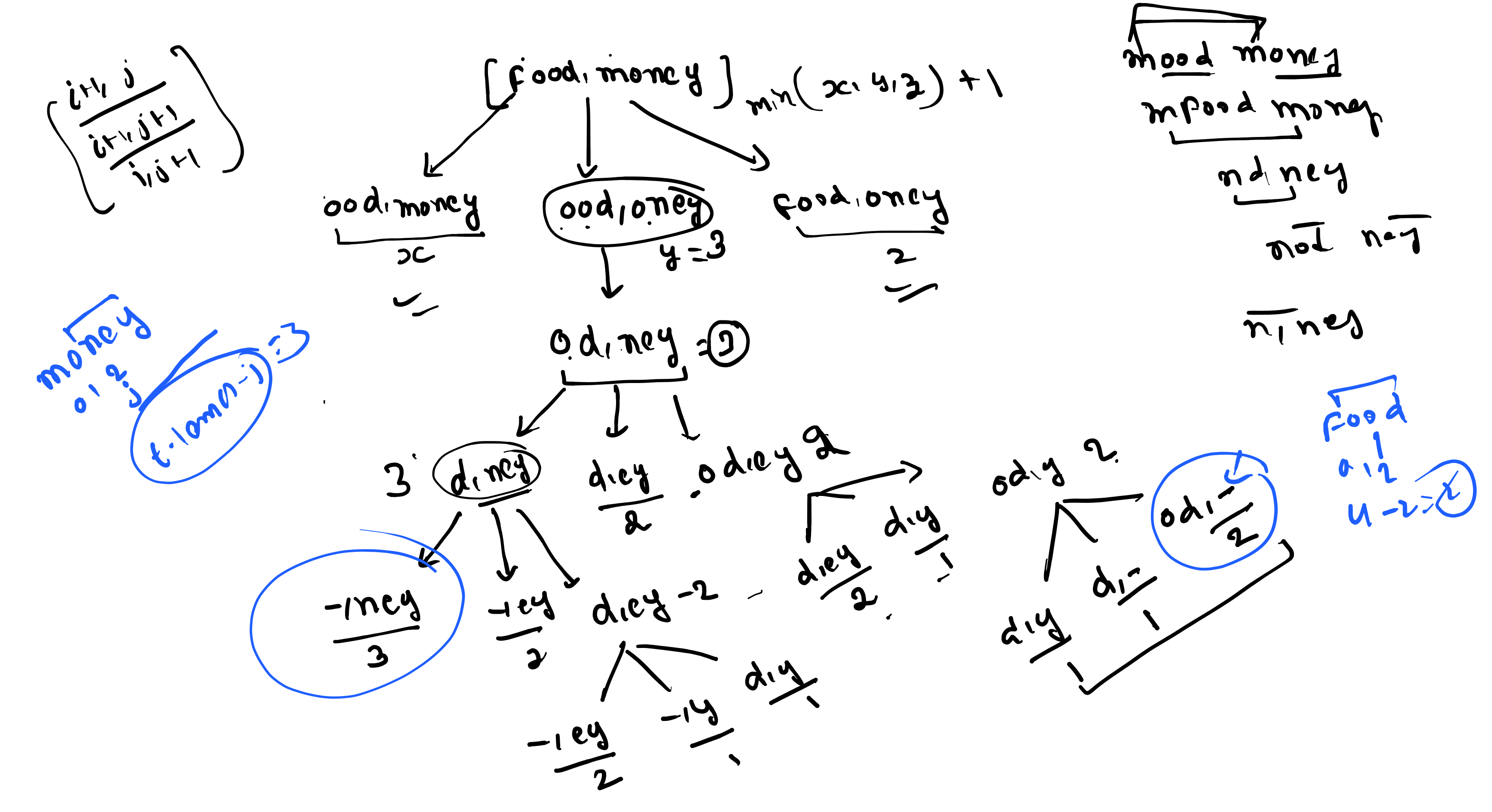
Handwritten text: $[1, 3, 7, 1, 7, 5]$ and $[1, 9, 2, 5, 1]$



	-	0	1	2	3	4	5
-	1	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	1	1	2	2	2	2	2
5	1	1	2	2	2	2	2



Handwritten text: $[1, 2, 5]$ and $[1, 2, 5] \rightarrow am$



You are packing for a vacation on the sea side and you are going to carry only one bag with capacity S ($1 \leq S \leq 1000$). You also have N ($1 \leq N \leq 1000$) items that you might want to take with you to the sea side. Unfortunately you can not fit all of them in the knapsack so you will have to choose. For each item you are given its size and its value. You want to maximize the total value of all the items you are going to bring. What is this maximum total value?

Handwritten text: $N=5$, $W=[1, 2, 3, 2, 2]$, $val=[8, 4, 0, 5, 3]$

